IGCSE 0607 Practice Questions — E10.8

Cumulative frequency diagrams

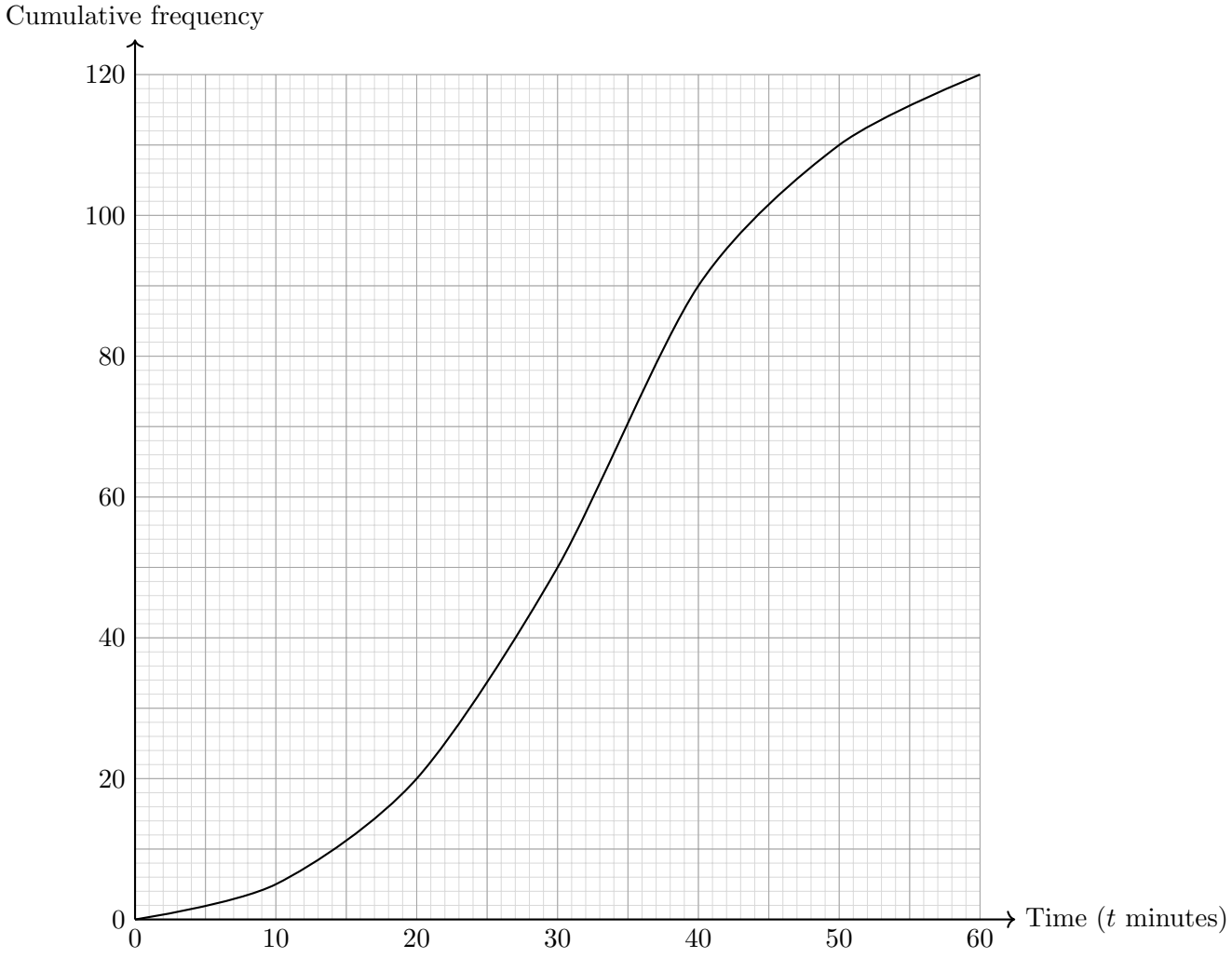
Total Questions: 17 — **Total Marks:** 127 Marks

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  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(13 Qs)*

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1 The cumulative frequency diagram shows the times taken, t minutes, by 120 students to complete a test.



(a) Use the diagram to estimate the median time.

..... minutes **[1]**

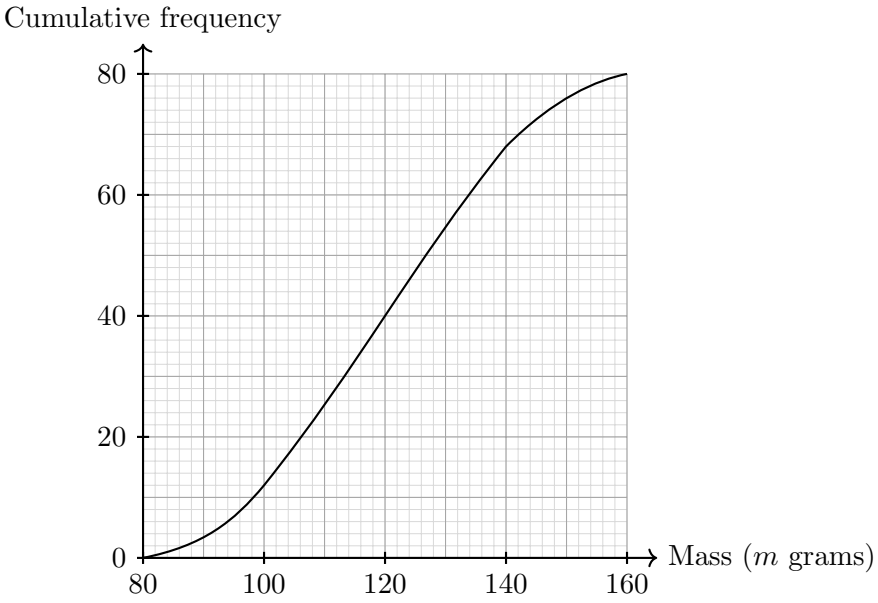
(b) Use the diagram to estimate the interquartile range.

..... minutes **[2]**

(c) Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

..... **[2]**

2 The cumulative frequency diagram shows information about the masses, m grams, of 80 apples.



(a) Use the cumulative frequency diagram to find an estimate for

(i) the median,

..... g **[1]**

(ii) the lower quartile,

..... g **[1]**

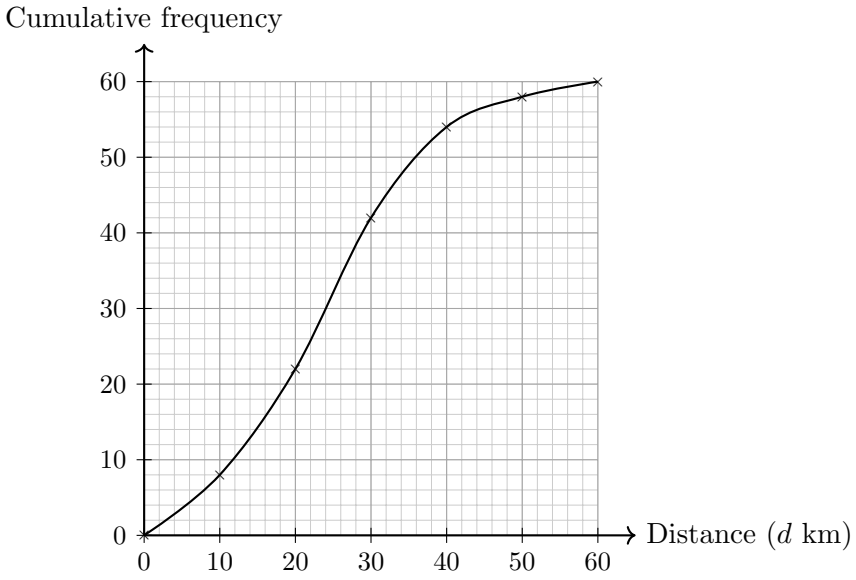
(iii) the interquartile range.

..... g **[2]**

(b) Find an estimate for the number of apples with a mass greater than 140 g.

..... **[1]**

3 The cumulative frequency diagram shows information about the distance, d km, travelled by 60 people to work.



(a) Use the diagram to find an estimate for:

(i) the median,

..... km **[1]**

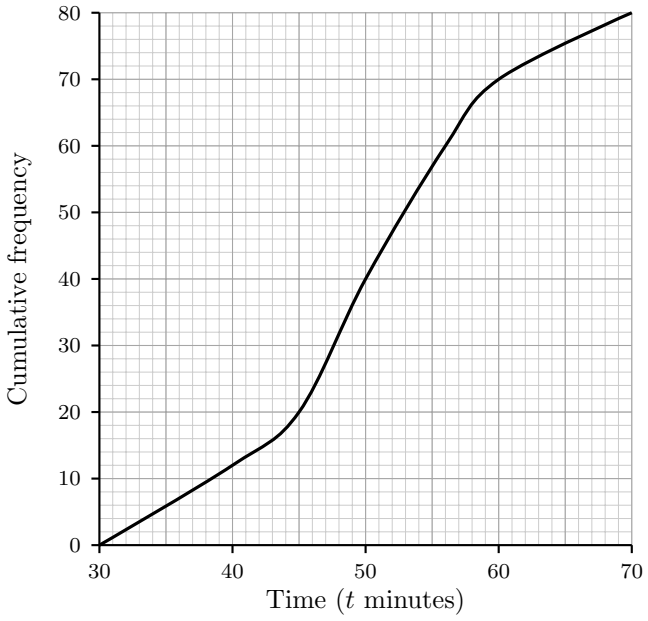
(ii) the interquartile range.

..... km **[2]**

(b) Find the number of people who travelled more than 45 km to work.

..... **[3]**

4 The cumulative frequency diagram shows information about the times, t minutes, taken by 80 runners to complete a 10 km race.



(a) Use the cumulative frequency diagram to find

(i) the median time,

..... minutes **[1]**

(ii) the lower quartile and the upper quartile,

Lower quartile = ... , Upper quartile = ... **[2]**

(iii) the interquartile range.

..... minutes **[1]**

(b) Find the number of runners who took more than 60 minutes to complete the race.

..... **[2]**

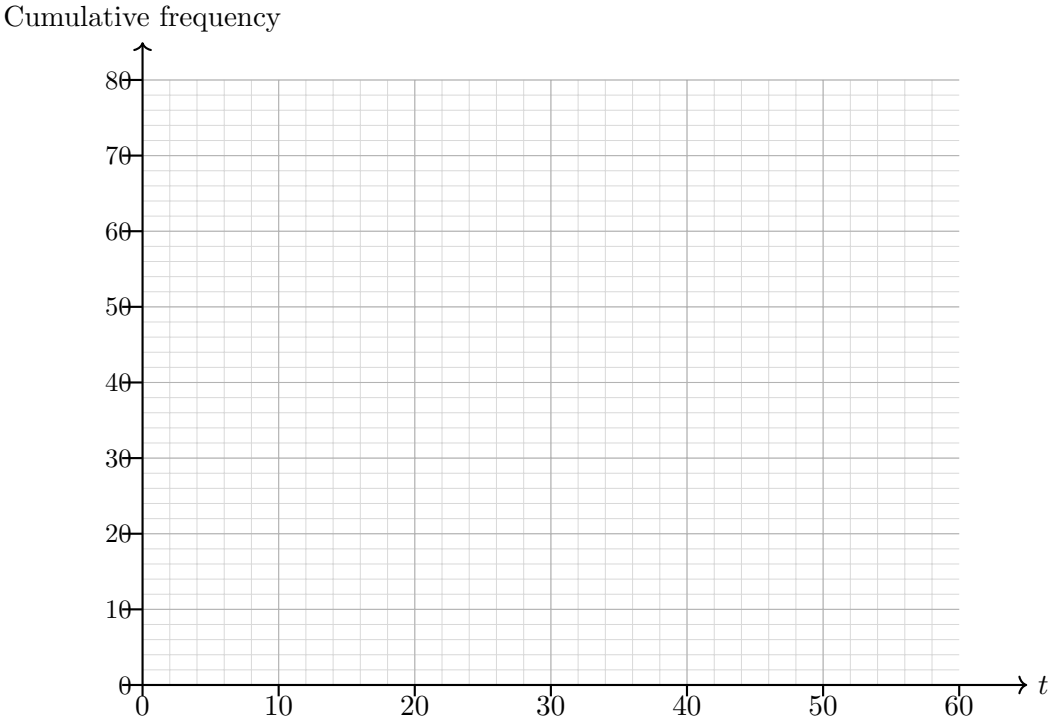
(c) Runners who finished in the fastest 15% of times qualified for a regional event.
Estimate the maximum qualifying time.

..... minutes **[2]**

5 The table shows the cumulative frequencies for the times, t minutes, taken by 80 students to complete a puzzle.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$	$t \leq 60$
Cumulative frequency	6	22	48	68	78	80

(a) On the grid, draw a cumulative frequency diagram to show this information.

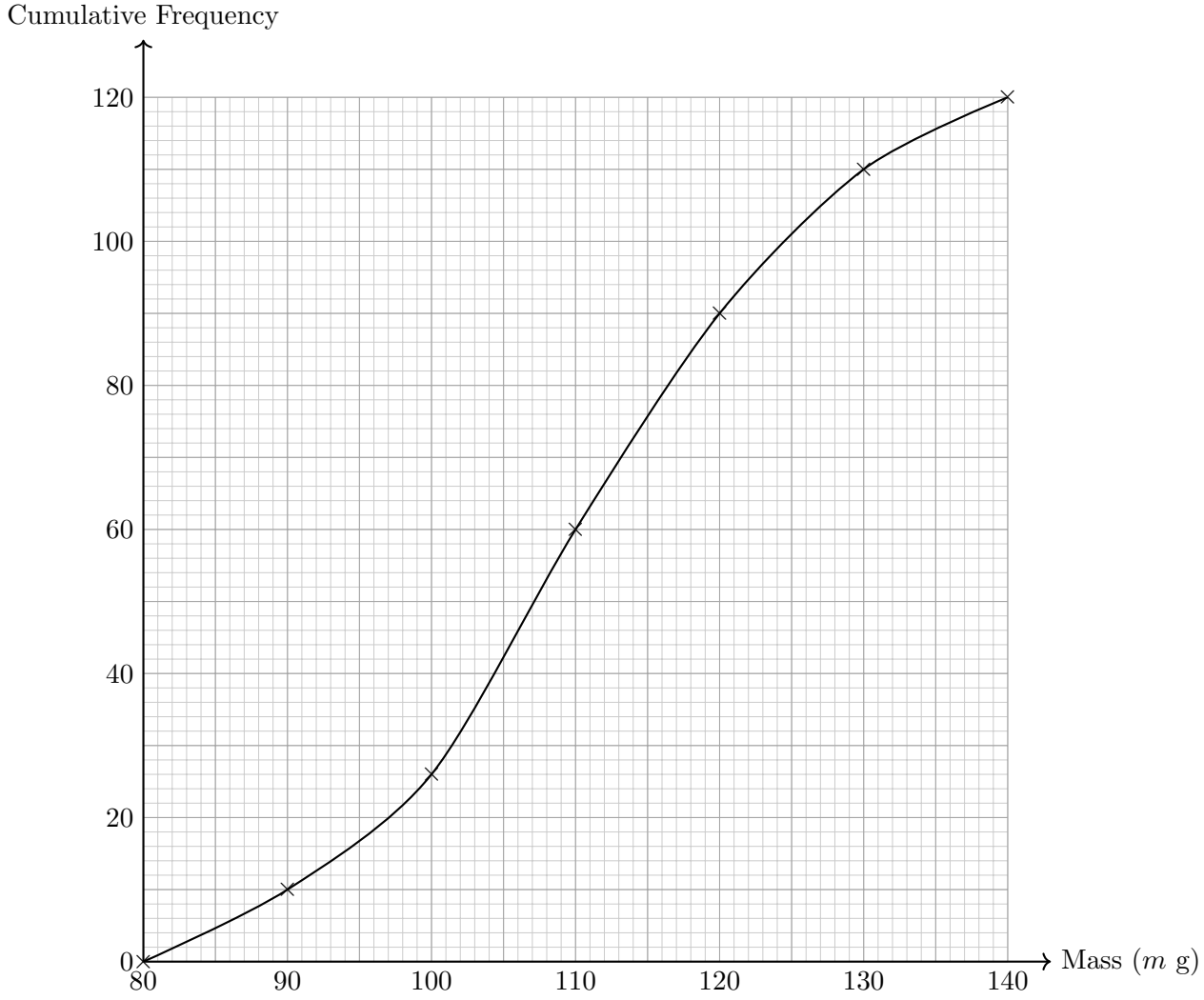


[3]

(b) Use your diagram to estimate the median time.

..... minutes **[2]**

6 The cumulative frequency diagram shows information about the masses, m grams, of 120 apples.



(a) Find the median mass.

..... g **[1]**

(b) Find the interquartile range.

..... g **[2]**

(c) Find the number of apples with a mass greater than 125 g.

..... **[2]**

7 The frequency table shows information about the times, t minutes, taken by 100 students to complete a puzzle.

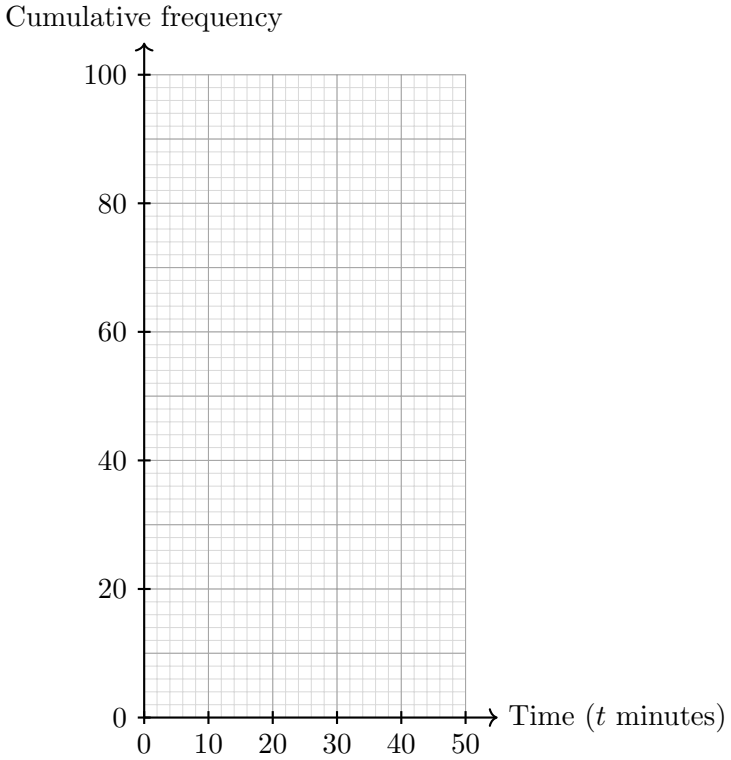
Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$
Frequency	12	24	36	18	10

(a) Complete the cumulative frequency table.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$
Cumulative frequency	12				100

[1]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your diagram to find an estimate for the 80th percentile.

..... minutes **[1]**

(d) Students who take more than 35 minutes must take a second test.
Use your diagram to estimate the percentage of students who must take a second test.

..... % **[2]**

8 The grouped frequency table shows the time, t minutes, taken by 80 students to complete a science test.

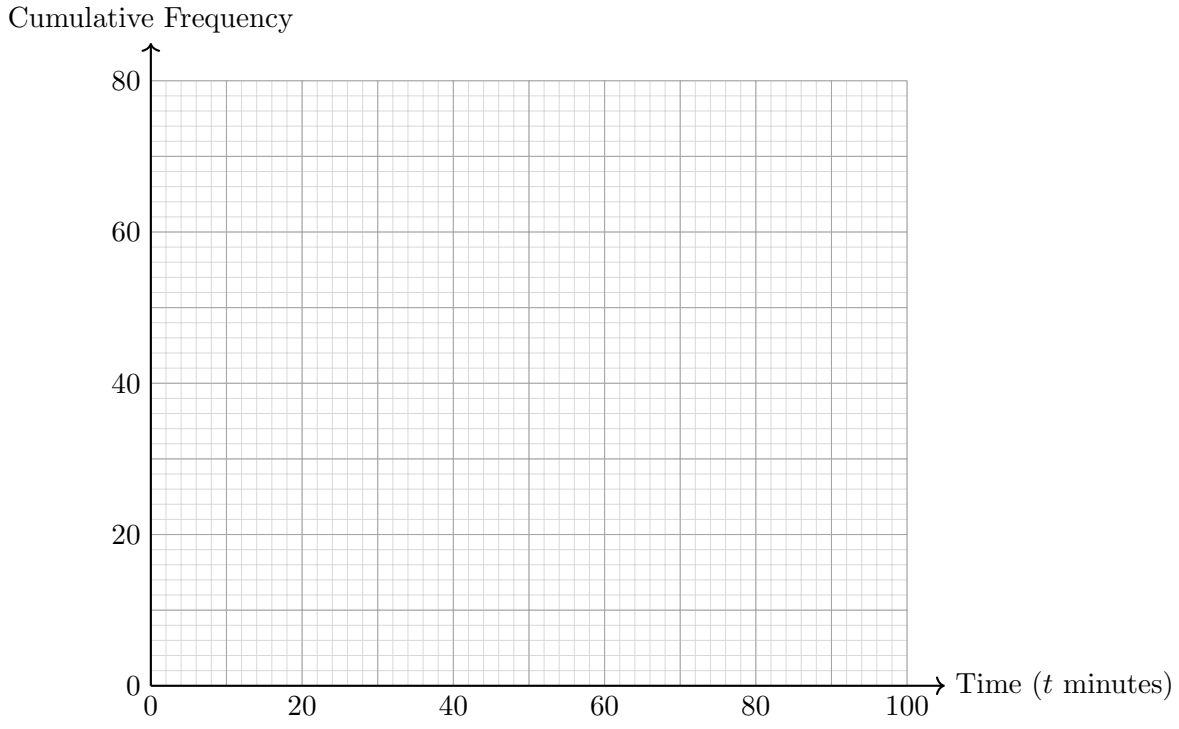
Time (t minutes)	$0 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$
Frequency	8	22	34	12	4

(a) Complete the cumulative frequency table.

Time (t minutes)	$t \leq 20$	$t \leq 40$	$t \leq 60$	$t \leq 80$	$t \leq 100$
Cumulative freq.	8				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your graph to estimate the number of students who took more than 55 minutes to complete the test.

..... **[2]**

9 A company records the salaries, s thousand dollars, of its 200 employees.
The frequency table shows the results.

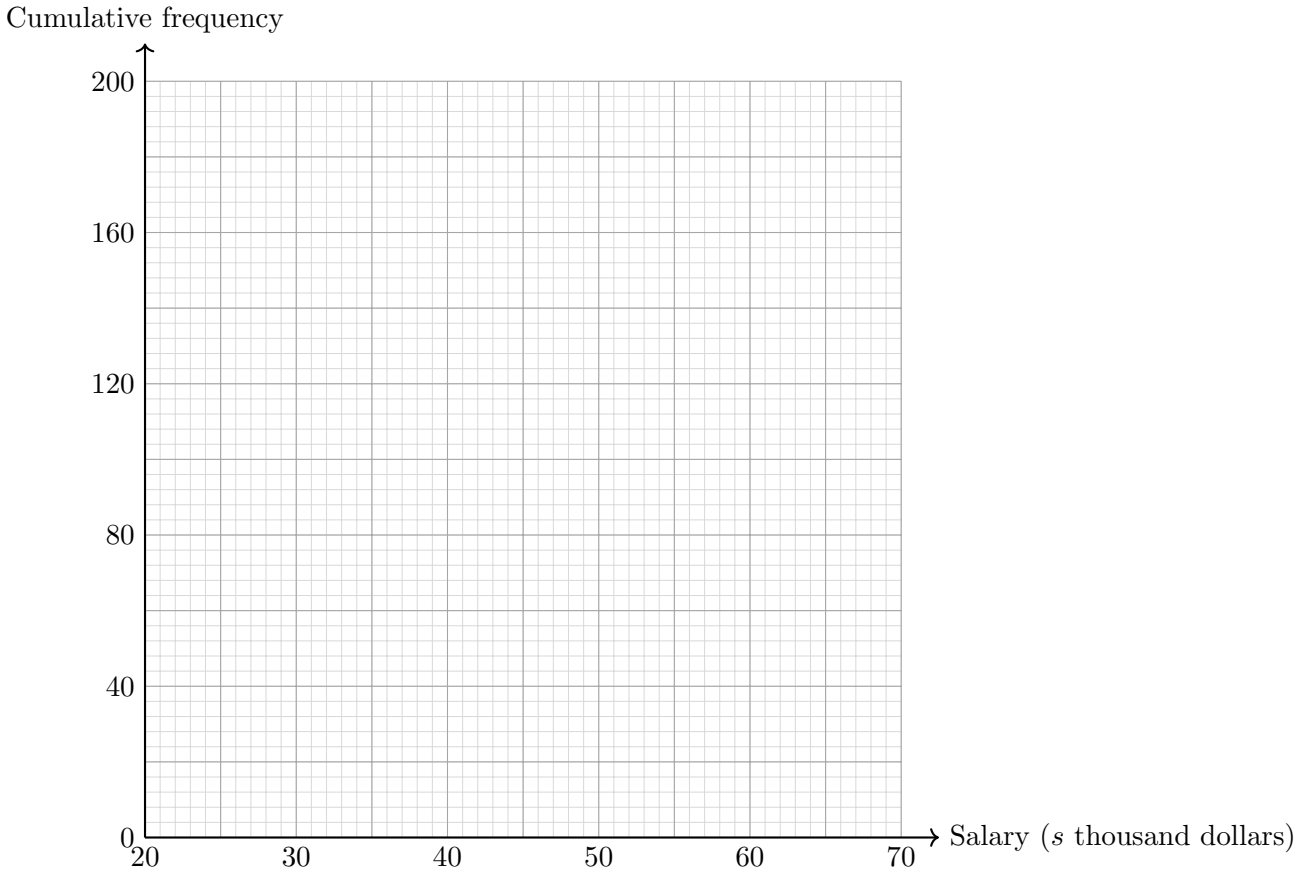
Salary (s thousand dollars)	$20 < s \leq 30$	$30 < s \leq 40$	$40 < s \leq 50$	$50 < s \leq 60$	$60 < s \leq 70$
Frequency	24	66	70	28	12

(a) Complete the cumulative frequency table.

Salary (s thousand dollars)	$s \leq 30$	$s \leq 40$	$s \leq 50$	$s \leq 60$	$s \leq 70$
Cumulative frequency	24				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.

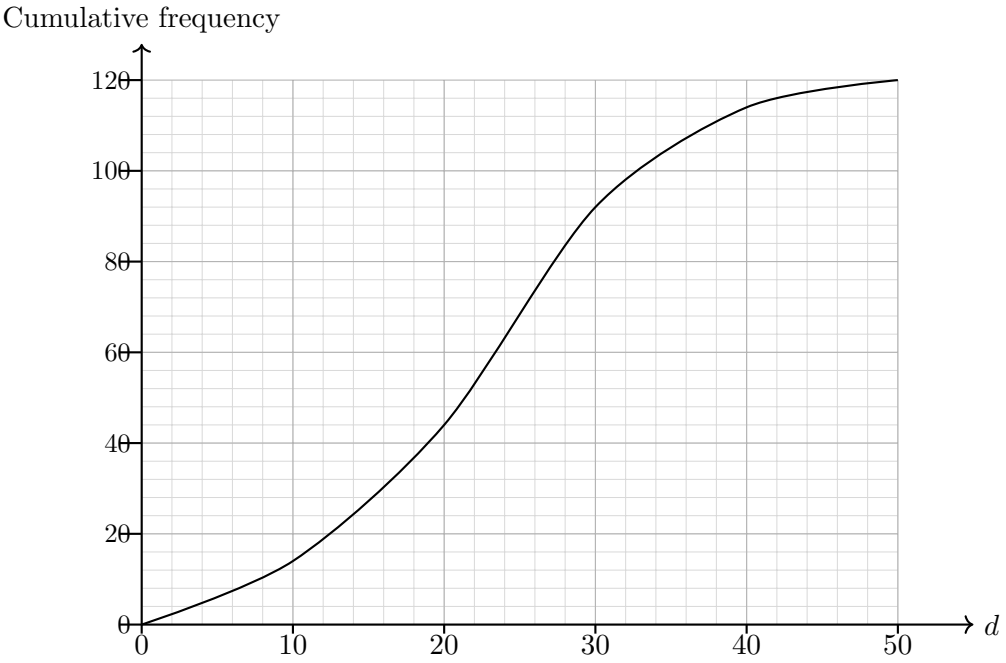


[3]

(c) Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

10 The cumulative frequency diagram shows information about the distances, d km, travelled by 120 people to work.



(a) Use the diagram to estimate the median distance.

..... km **[2]**

(b) Use the diagram to estimate the interquartile range.

..... km **[3]**

(c) Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

11 A table shows information about the masses of 120 apples.

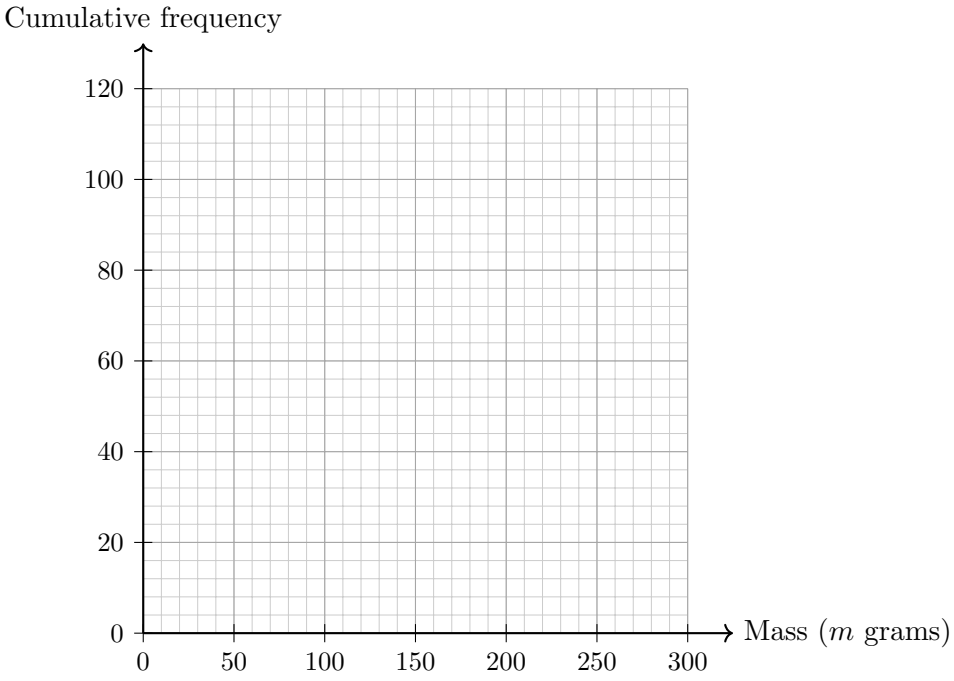
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	12	34	46	20	8

(a) Complete the cumulative frequency table.

Mass (m grams)	$m \leq 100$	$m \leq 150$	$m \leq 200$	$m \leq 250$	$m \leq 300$
Cumulative frequency	12				120

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your diagram to estimate the 60th percentile.

..... g **[2]**

(d) Apples with a mass greater than 220 grams are classified as large.
Estimate the number of large apples.

..... **[3]**

12 A supermarket records the checkout waiting time, w seconds, for 120 customers.
The grouped frequency table shows the results.

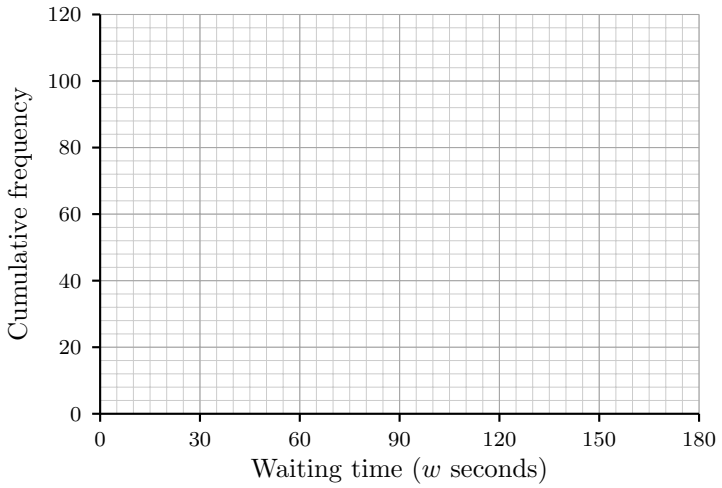
Waiting time (w s)	Frequency
$0 < w \leq 30$	10
$30 < w \leq 60$	22
$60 < w \leq 90$	44
$90 < w \leq 120$	28
$120 < w \leq 150$	12
$150 < w \leq 180$	4

(a) Complete the cumulative frequency table below.

Waiting time (w s)	Cumulative frequency
$w \leq 30$	10
$w \leq 60$	
$w \leq 90$	
$w \leq 120$	
$w \leq 150$	
$w \leq 180$	

[2]

(b) On the grid below, draw a cumulative frequency diagram to represent this information.
Mark your points clearly with small crosses ($\times$) and join them with a smooth curve.



[3]

(c) Use your cumulative frequency diagram to estimate

(i) the median waiting time,

..... seconds **[1]**

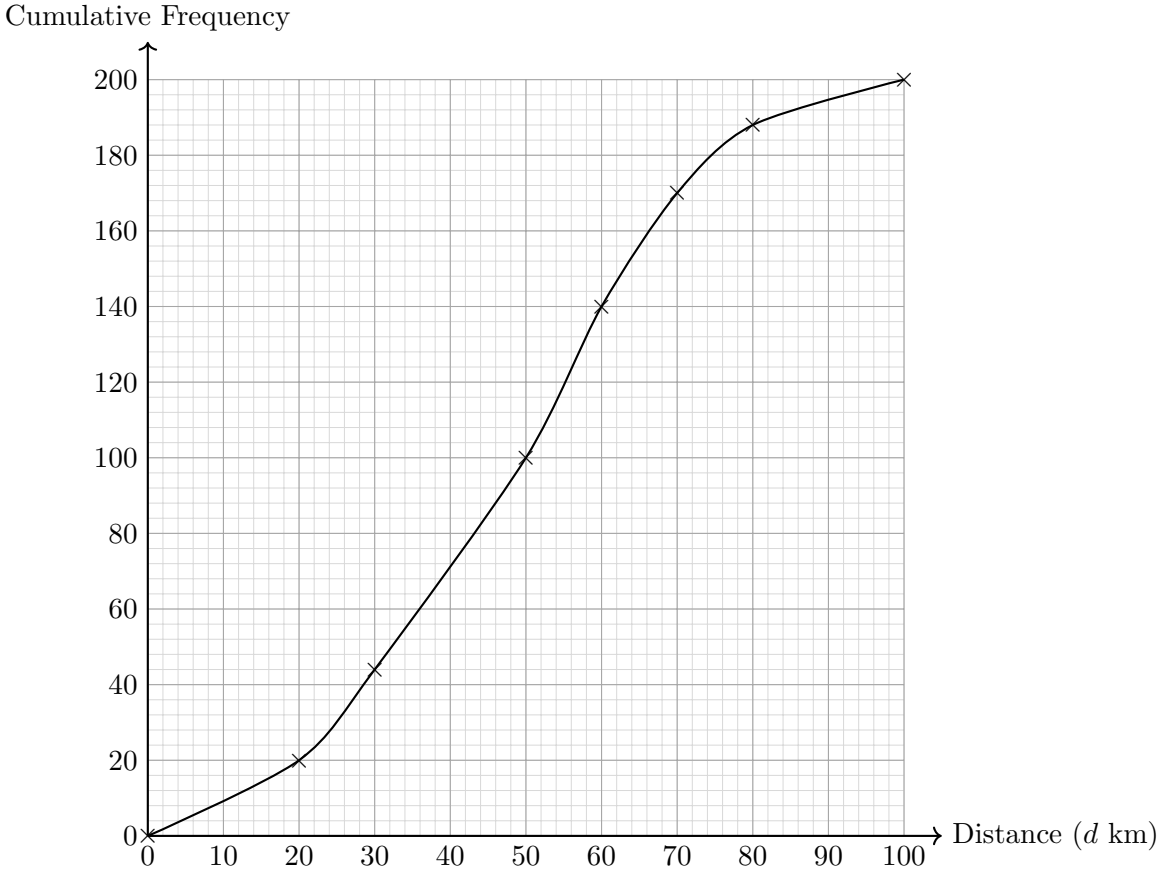
(ii) the 80th percentile.

..... seconds **[2]**

(d) The supermarket manager sets a target that at least 85% of customers must wait no more than 115 seconds.
Use your diagram to determine whether this target is met, showing your working clearly.

..... **[3]**

13 The cumulative frequency diagram shows information about the distance, d kilometres, travelled by 200 delivery vans in a single day.



(a) Find an estimate for the 60th percentile.

..... km **[2]**

(b) 15% of the vans travelled further than k km.
Find the value of k.

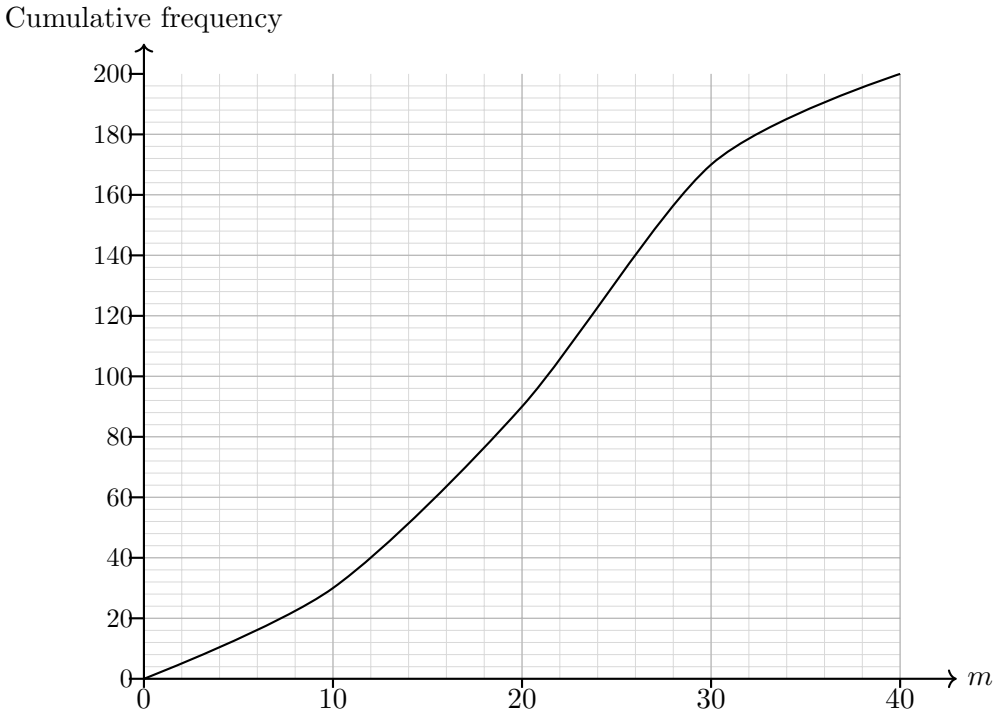
$k =$ ..... **[2]**

(c) Use the cumulative frequency diagram to complete the frequency table.

Distance (d km)	$0 < d \leq 30$	$30 < d \leq 50$	$50 < d \leq 80$	$80 < d \leq 100$
Frequency				

[3]

14 The cumulative frequency diagram shows information about the masses, m grams, of 200 parcels.



(a) Find an estimate for the 80th percentile.

..... grams **[2]**

(b) Use the cumulative frequency diagram to complete the frequency table.

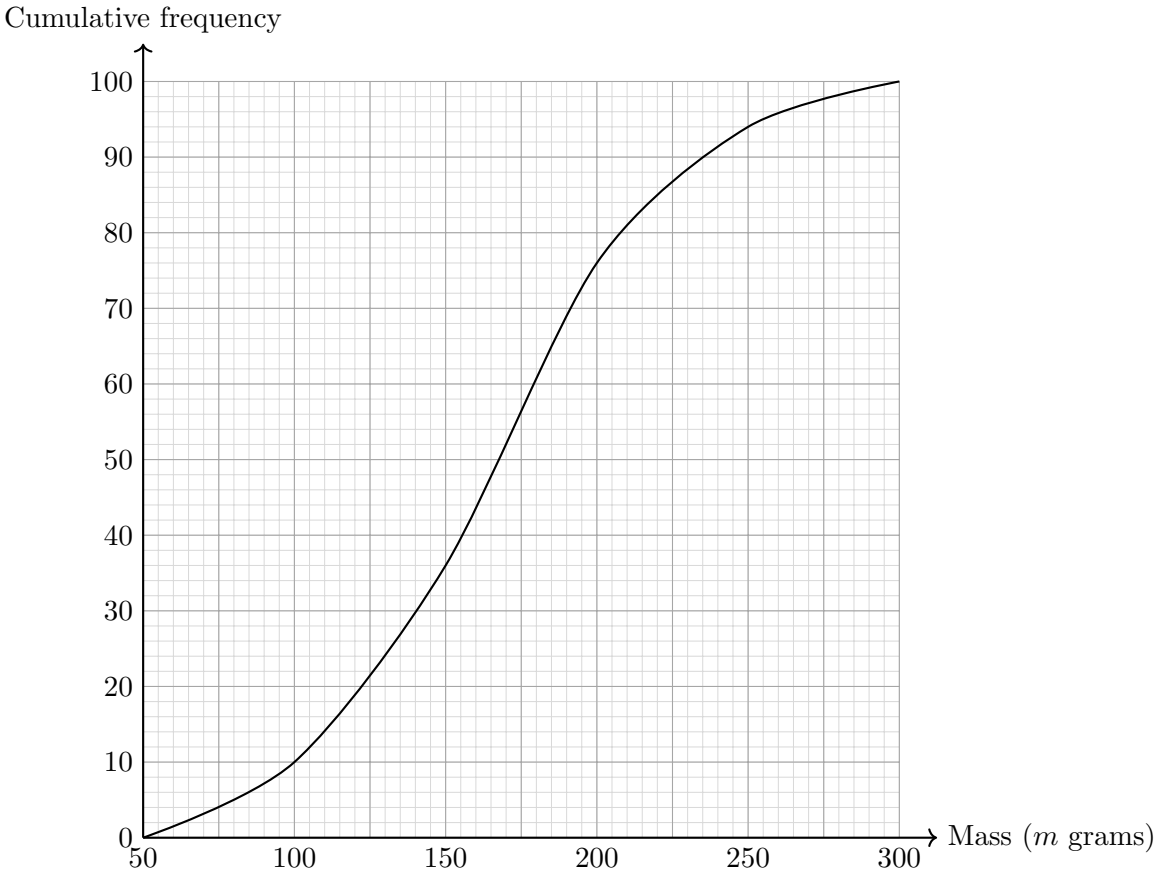
Mass (m grams)	$0 < m \leq 10$	$10 < m \leq 20$	$20 < m \leq 30$	$30 < m \leq 40$
Frequency				

[2]

(c) Calculate an estimate of the mean mass.

..... grams **[4]**

15 The cumulative frequency curve shows information about the masses, m grams, of 100 potatoes.



(a) Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

(b) The curve was drawn using grouped data.
Find the values of p, q, and r.

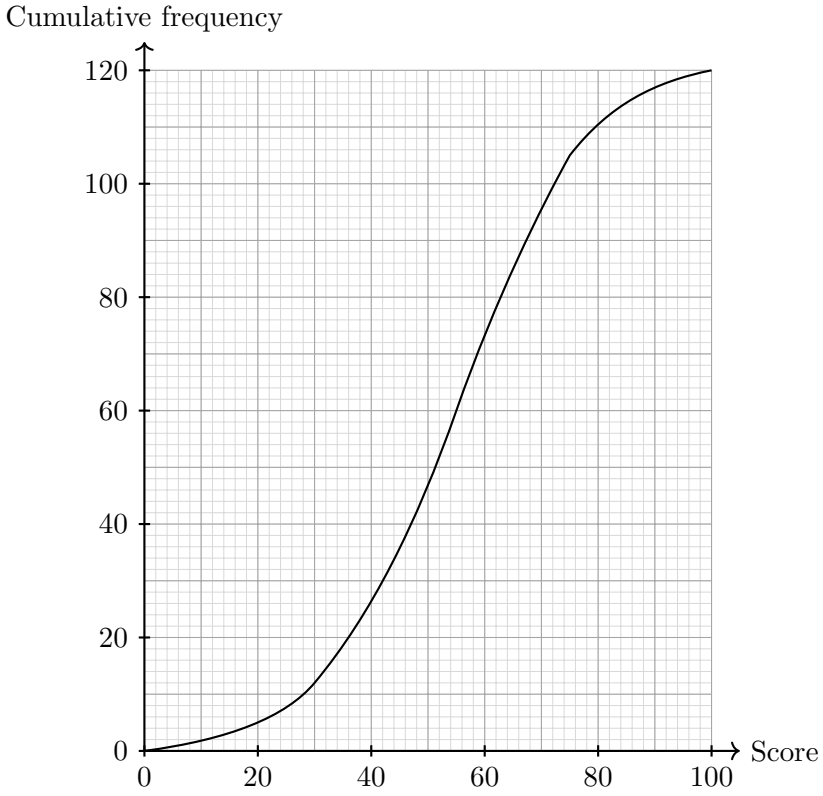
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	p	q	40	r	6

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

(c) Calculate an estimate of the mean mass of the potatoes.

..... g **[3]**

16 The cumulative frequency diagram shows information about the test scores of 120 students in Mathematics.
The maximum possible score is 100.



(a) Find an estimate for the interquartile range of the Mathematics scores.

..... **[2]**

(b) The top 15% of students are awarded a distinction.
Use the diagram to estimate the minimum score needed for a distinction.

..... **[2]**

(c) The same 120 students also took a Physics test.
For the Physics scores, the median is 62 and the interquartile range is 16.
Make two comparisons between the Mathematics scores and the Physics scores.

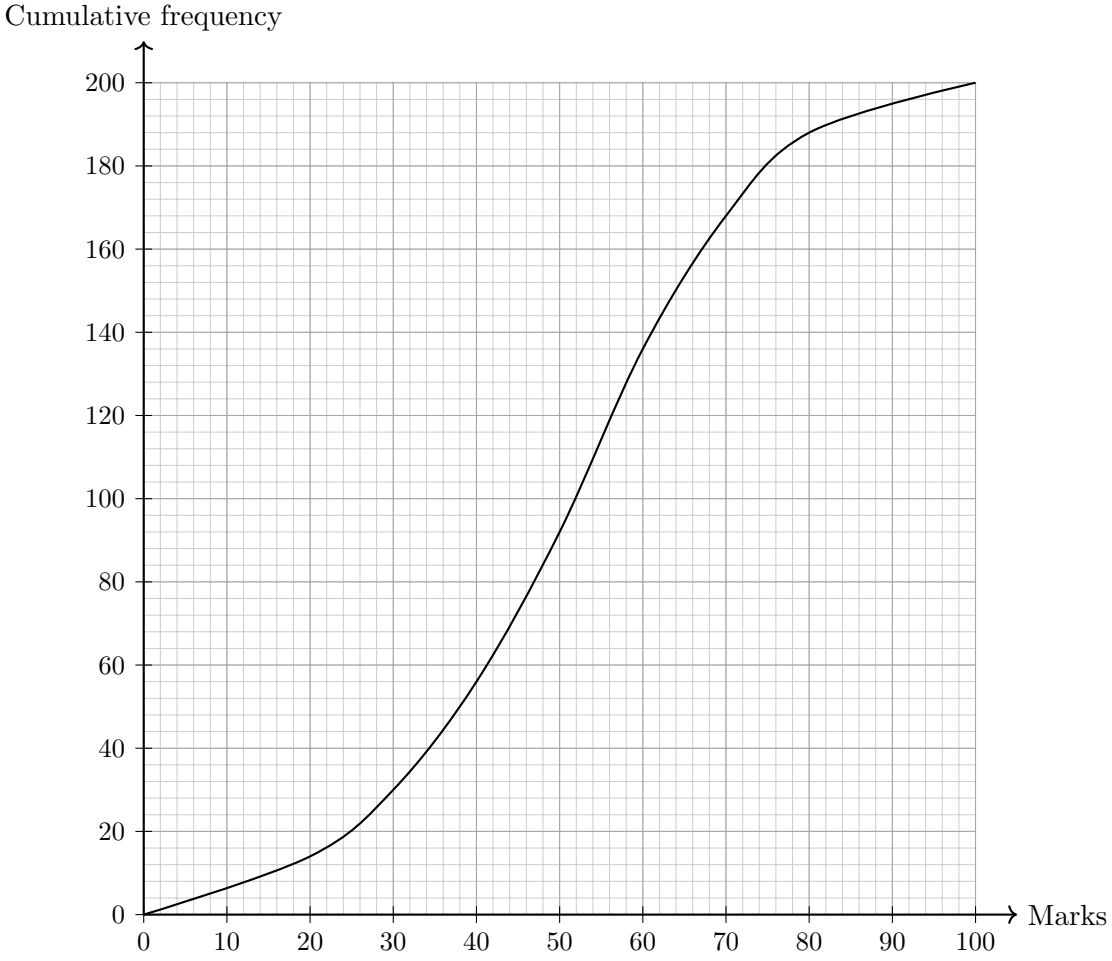
.....

..... **[2]**

(d) Two of the 120 students are chosen at random.
Using the cumulative frequency diagram, calculate an estimate for the probability that both students scored more than 60 in Mathematics.

..... **[3]**

17 The cumulative frequency diagram shows the distribution of marks scored by 200 candidates in an examination.



(a) Use your graph to estimate the interquartile range.

..... **[2]**

(b) 15% of the candidates score less than m marks.
Find the value of m.

$m =$ ..... **[2]**

(c) Candidates scoring more than P marks are awarded a Distinction.
Given that 30 candidates are awarded a Distinction, find the value of P.

$P =$ ..... **[2]**

(d) Candidates who score less than 45 marks have to take a supplementary test.
Calculate the percentage of candidates who do NOT have to take this test.

..... % **[4]**